

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
First Semester of B.Sc. Physics (Theory) Examination November 2018
PD101 Mathematical Physics-I

Date: 19/11/2018, Monday Time: 10:00 a.m. to 10:30 a.m. Maximum Marks: 20

MCQ

Important Instructions:

1. Tick the correct answer and it should be written in question paper itself.
 2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
 3. Each question carries one mark.
 4. All notations are standard.
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Q - I Choose the most appropriate answer from the given options for the followings:

1. The degree and order of the differential equation

$$\sqrt[3]{\left(\frac{dy}{dx}\right)^2 + \left(\frac{d^2y}{dx^2}\right)^3} = \frac{d^4y}{dx^4} \text{ are respectively } \underline{\hspace{2cm}}.$$

- (a) 4 and 3 (c) 3 and 2
(b) 3 and 4 (d) None of these

2. The set $\{e^x, e^{-x}\}$ is _____.

- (a) both linearly independent and linearly dependent (c) Linearly Independent
(b) Linearly Dependent (d) None of these

3. The function $f(x, y, z) = \frac{(xy)^2}{\sqrt{x+y}}$ is a homogeneous function of degree _____.

- (a) $\frac{7}{2}$ (c) $\frac{5}{2}$
(b) $\frac{3}{2}$ (d) None of these

4. The Maclaurin's expansion of the function $f(x) = e^{x-1}$ is _____.

- (a) $e \left[1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right]$ (c) $\left[1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots \right]$
(b) $\left[1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right]$ (d) $\frac{1}{e} \left[1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right]$

5. The Jacobian $\frac{\partial(u, v)}{\partial(x, y)}$ of the functions $u = x^3 - y^3$ and $v = x^2 - y^2$ is _____.

- (a) $6(xy^2 - x^2y)$ (c) $6xy(x + y)$
(b) $6(x^2y - xy^2)$ (d) None of these

6. If $m = \pm 4i$, $\pm 4i$ are the roots of the auxiliary equation of the differential equation then the general solution of the differential equation is _____.

- (a) $e^{4x}\{(c_1 + c_2) \cos 4x + (c_3 + c_4) \sin 4x\}$
(b) $e^{4x}\{(c_1 + c_2x) \cos 4x + (c_3 + c_4x) \sin 4x\}$
(c) $(c_1 + c_2x) \cos 4x + (c_3 + c_4x) \sin 4x$
(d) None of these

7. The matrix of transformation of vectors under rotation is _____.

- (a) $\begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ (c) $\begin{bmatrix} \cos \theta & -\sin \theta \\ \cos \theta & \sin \theta \end{bmatrix}$
- (b) $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ (d) None of these

8. The Wronskian $W(\log x, \log x^n) =$ _____.

- (a) 1 (b) n (c) $\log x$ (d) 0

9. If $\vec{F}$ represents the velocity of a liquid then flux is represented as _____.

- (a) $\iint_S \vec{F} \cdot d\vec{S}$ (c) $\iiint_S \text{div} \vec{F} dS$
- (b) $\iint_S (\vec{F} \cdot \hat{n}) dS$ (d) $\iint_S \vec{F} dx dy$

10. $\int_0^1 \int_0^5 \int_0^{2e} dy dx dz =$ _____.

- (a) $2e + 5$ (b) $2e - 5$ (c) $10e$ (d) $2e$

11. The general solution of the linear differential equation $\frac{dy}{dx} + P(x)y = Q(x)$ is _____.

- (a) $xe^{\int P dx} = \int Q(x)e^{\int P dx} dx + C$
- (b) $e^{\int P dx} = \int Q(x)e^{\int P dx} dx + C$
- (c) $ye^{\int P dx} = \int Q(x)e^{\int P dx} dx + Ce^{-\int P dx}$
- (d) $y = e^{-\int P dx} \int Q(x)e^{\int P dx} dx + Ce^{-\int P dx}$

12. The vector $\vec{F} = (x + 3y)\hat{i} + (y - 3z)\hat{j} + (x - 2z)\hat{k}$ is _____.

- (a) not irrotational (c) irrotational
- (b) not solenoidal (d) both irrotational and solenoidal

13. The unit normal to $x^2 + y^2 + z^2 = 5$ at $(0, 1, 2)$ is _____.

- (a) $\frac{1}{\sqrt{5}}(\hat{i} + \hat{j} + \hat{k})$ (c) $\frac{1}{\sqrt{5}}(\hat{i} - \hat{j} + \hat{k})$
- (b) $\frac{1}{\sqrt{5}}(\hat{i} + \hat{j} - \hat{k})$ (d) $\frac{1}{\sqrt{5}}(\hat{j} + 2\hat{k})$

14. Mean, median and mode all are equal for _____ distribution.
- (a) normal (c) binomial
- (b) Poisson (d) None of these
15. If the events A and B are mutually exclusive and exhaustive then $P(A \cup B) =$ _____.
- (a) 0 (c) 1
- (b) $P(A \cap B)$ (d) None of these
16. If δ denotes the dirac delta function then $\int_0^2 (x^2 + 4)\delta(x - 5)dx =$ _____.
- (a) $\frac{32}{3}$ (c) ∞
- (b) 0 (d) None of these
17. $\int_b^c \delta(x - a)dx =$ _____, if $b < a < c$, where δ denotes the dirac delta function.
- (a) 0 (b) ∞ (c) 1 (d) $\frac{1}{2}$
18. The probability that an odd number appear in a single toss of a fair die is _____.
- (a) $\frac{1}{3}$ (b) $\frac{1}{6}$ (c) $\frac{5}{6}$ (d) $\frac{1}{2}$
19. For a binomial distribution, if n is the total number of trials then, $\sum_{x=0}^n p^x (1 - p)^{n-x} =$ _____.
- (a) 0 (c) 1
- (b) $\frac{1}{2}$ (d) None of these
20. For the Poisson distribution, _____ is true.
- (a) the number of trials n should be infinitely large
- (b) the probability of successes p for each trials should be very large
- (c) the distribution constant $\lambda = np$ should be infinite
- (d) None of these

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
First Semester of B.Sc. Physics (Theory) Examination November 2018
PD101 Mathematical Physics-I

Date: 19/11/2018, Monday Time: 10:30 a.m. to 01:00 p.m. Maximum Marks: 50

Instructions:

1. **Section I and Section II must be written in separate answer books..**
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Figures to the right indicate marks.
4. All notations are standard.

SECTION I

Q - II (A) Attempt any **four** of the followings: **[8]**

- (1) Evaluate $\lim_{x \rightarrow \frac{\pi}{2}} (\cos x)^{\cos x}$.
- (2) Solve $(2x + y + 1)dx + (x + 3y + 2)dy = 0, y(0) = 0$.
- (3) Solve $\frac{dy}{dx} + \frac{4x}{x^2 + 1}y = \frac{1}{(x^2 + 1)^3}$.
- (4) Find the value of m so that the vectors $2\hat{i} - 4\hat{j} + 5\hat{k}$, $\hat{i} - m\hat{j} + \hat{k}$, and $3\hat{i} + 2\hat{j} - 5\hat{k}$ are coplanar.
- (5) Prove that $\vec{F} = (y^2 - z^2 + 3yz - 2x)\hat{i} + (3xz + 2xy)\hat{j} + (3xy - 2xz + 2z)\hat{k}$ is solenoidal.
- (6) A car-hire firm has two cars, which it hires out day by day. The number of demands for a car on each day is distributed as a Poisson distribution with a mean of 1.5. Calculate the proportion of days on which neither car is used.

(B) Attempt any **four** of the followings: **[12]**

- (1) Find the points of discontinuity for the following function and also draw its graph:

$$f(x) = \begin{cases} x, & \text{when } 0 \leq x < 1 \\ 2, & \text{when } x = 1 \\ x + 1, & \text{when } 1 < x \leq 2 \end{cases}$$

- (2) Find the directional derivative of $\phi(x, y, z) = x^2yz + 4xz^2$ at $(1, -2, 1)$ in the direction of $2\hat{i} - \hat{j} - 2\hat{k}$.
- (3) Solve $y'' - 9y = 0, y(0) = 2, y'(0) = -1$.
- (4) Solve $x^4 \frac{dy}{dx} + x^3y = -\sec(xy)$.
- (5) Using Green's theorem, evaluate $\int_C x^2y dx + x^2 dy$, where C is the boundary described under clockwise of the triangle with vertices $(0,0)$, $(1,0)$ and $(1,1)$.

- (6) Suppose that 5 men out of 100 and 25 women out of 10000 are colorblind. A colorblind person is chosen at random. What is the probability of his being male? (Assume males and females to be equal in number.)

SECTION II

Q - III (A) Attempt any **three** of the following. [12]

- (1) Solve $\frac{d^2y}{dx^2} + 4y = \tan 2x$ using method of variation of parameters.
- (2) Find the maximum value of $u = x^p y^q z^r$ when the variables x, y, z are subject to the condition $ax + by + cz = p + q + r$.
- (3) Calculate the approximate value of $\sqrt{10}$ to four decimal places by taking first four terms of an appropriate Taylor's expansion.
- (4) Prove that in spherical polar co-ordinate system, $\vec{T}_r, \vec{T}_\theta, \vec{T}_\phi$ are mutually perpendicular.
- (5) By assuming that the probability of a child being a boy or a girl is equal, find the number of families out of 400 consisting 3 children each having (i) all boys (ii) at most one boy.

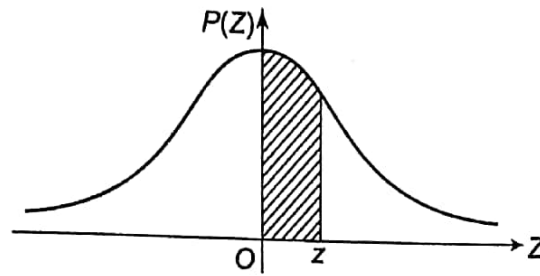
(B) Attempt any **three** of the following. [18]

- (1) Show that the vector field $\vec{F} = 2x(y^2 + z^3)\hat{i} + 2x^2y\hat{j} + 3x^2z^2\hat{k}$ is conservative. Find its scalar potential and work done in a moving particle from $(-1, 2, 1)$ to $(2, 3, 4)$.
- (2) Use Stoke's theorem to evaluate $\oint_C \vec{F} \cdot d\vec{r}$ if $\vec{F} = (x + y)\hat{i} + (2x - z)\hat{j} + (y + z)\hat{k}$ and C is the boundary of the triangle $(2, 0, 0)$, $(0, 3, 0)$ and $(0, 0, 6)$.
- (3) Let $F(u, v, w)$ be a vector point function of the orthogonal curvilinear co-ordinates u, v, w . Then prove that for $\vec{F} = (f_1, f_2, f_3)$,

$$\text{div}.\vec{F} = \nabla.\vec{F} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u}(h_2 h_3 f_1) + \frac{\partial}{\partial v}(h_3 h_1 f_2) + \frac{\partial}{\partial w}(h_1 h_2 f_3) \right].$$

- (4) Verify Gauss' divergence theorem for $\vec{F} = (x^2 - yz)\hat{i} + (y^2 - zx)\hat{j} + (z^2 - xy)\hat{k}$ over the surface of the rectangular parallelepiped $0 \leq x \leq a, 0 \leq y \leq b, 0 \leq z \leq c$.
- (5) (i) A manufacturer knows from his experience that the resistances of resistors he produces is normal with $\mu = 100 \text{ ohms}$ and $\text{SD} = \sigma = 2 \text{ ohms}$. What percentage of resistors will have resistances between 98 ohms and 102 ohms ?
(ii) The average seasonal rainfall in a place is 16 inches with a SD of 4 inches. What is the probability that the rainfall in that place will be between 20 and 24 inches in a year? (**Standard normal table: P.T.O.**)

Standard Normal (Z) Table, Area between 0 and z



Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3990	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4115	0.4131	0.4147	0.4162
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990